University of Science and Technology Houari Boumediene
5th Year - Software Engineering
2026-2027

Game Development: Theory and Practice
Practical Project
Design and Development of a Video Game

1. Project Objective

The objective of this project is to design and develop a video game by applying the theoretical and practical concepts covered in the course.

Students must propose an original game concept, produce a complete Game Design Document (GDD) describing the overall game, and implement one complete, playable level.

2. Project Organization

- **Team size:** 2–4 students.
- **Deliverables:** Game Design Document, source code, and final poster presentation.
- **Scope:** The GDD must describe the complete proposed game, but only one level must be designed and implemented.
- **Deadline:**
  - Groups: 08/11/ 2026
  - GDD: 22/11/2026
  - Final Product: January

3. Part I — Game Design Document (GDD)

Students must prepare a Game Design Document describing the general idea and design of their proposed game. The GDD must cover the following sections.

1. Game Overview
2. Story and Game World
3. Gameplay and Mechanics
4. Game Structure and Progression
5. Visual and Audio Design
6. Technical Design
7. References and Inspirations

4. Part II — Design and Implementation of One Level

Students must design and implement **one complete, playable level** of their proposed game.

Students may choose one of the following game formats:

A. 2D game

- Sprites, tiles, or 2D graphics.
- Keyboard input and game loop.
- Collision detection.
- Movement and appropriate physics.
- Animation and win/lose conditions.

B. 2.5D game

- A scene with an appropriate camera and perspective.
- Objects, materials, lighting, and scene composition.
- Player or object interaction.
- Collision detection and physics where appropriate.
- A playable objective and animated interaction.

C. 3D game

- GameObjects, components, and appropriate materials.
- Colliders and Rigidbody or another justified physics approach.
- Animation or animated interactions.
- A complete playable level with a clear objective.

5. Deliverables

Each team must submit the following:

1. **Game Design Document**
2. **Report**
   A brief technical report describing:
   - The chosen game concept and format.
   - The development environment and tools.
   - The main implemented features (functions).
   - AI prompts if AI generated code was used.
   - The physics, collision, animation, and other relevant technical concepts used.
   - The main difficulties encountered and how they were addressed.
   - The division of work between team members.
3. **Final Demonstration**
   Each team must demonstrate the game in class and explain its design decisions and implementation.

General Requirements

- The GDD must describe the overall intended game, while implementation is limited to one level.
- The selected level must be complete and playable, rather than a collection of disconnected scenes.
- External assets and libraries may be used, but their sources must be acknowledged.
- If AI is used, include prompts in the technical report.
- Teams must ensure that all members contribute to the project.
- Students must be prepared to explain and justify their design and implementation choices.
- Originality, coherent game design, functionality, and appropriate technical implementation will be considered during evaluation.